Axles and alignment

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use measurements and observations to improve straight rolling.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 18.*
35 J; heating or sound are possible.

02. Worked example: explain the model
Parallel axle lines, perpendicular intended travel, positive wheel clearance; actual dimensions recorded when measured.

03. Read and interpret the model
Student marks measurable alignment and free clearance.

04. Apply the physics
Forward distance 100 cm; lateral offset +8 cm right. Use repeated actual measurements.

05. Explain your reasoning
No; wheel rubbing, unequal wheel sizes, surface slope or push could matter. Inspect and repeat controlled trials.

06. Review: connect an earlier idea *Review: Unit 7, Day 11.*
$E=Pt=24$ J.

07. Exit check
Specific recorded change with a measurable outcome such as reduced offset or freer rolling.

08. STRETCH: test the model
Possibilities unequal wheel drag, skewed axles, unequal effective wheel diameters; unpowered roll/spin checks and one-variable changes distinguish them.

Axles and alignment

antocciphysics | TEACHER KEY | HONORS + MATCHING SEI

Learning target: Use measurements and observations to improve straight rolling.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 18.*
$|emf| = N|\Delta\Phi|/\Delta t = 0.7$ V.

02. Worked example: explain the model
Parallel axle lines, perpendicular intended travel, positive wheel clearance; actual dimensions recorded when measured.

03. Read and interpret the model
Student marks measurable alignment and free clearance.

04. Apply the physics
3.43° to the side; magnitude 2.004 m.

05. Explain your reasoning
No; wheel rubbing, unequal wheel sizes, surface slope or push could matter. Inspect and repeat controlled trials.

06. Review: connect an earlier idea *Review: Unit 7, Day 11.*
$V = 24$ V; $P = IV = I^2R = 144$ W.

07. Exit check
Before: mean 0.20 m right, range 0.18-0.22 m (width 0.04 m). After: mean 0.05 m right, range 0.04-0.06 m (width 0.02 m). All after-trial offsets are smaller and less spread, supporting improvement under matched conditions.

08. STRETCH: test the model
Mean 0.22 m left; range 0.20-0.24 m. Change one alignment variable; keep track/release and trial count fixed, retain all results.